Week 8: Bayesian statistical inference

F79MA Statistical Models A

Heriot-Watt University

Bayesian Inference

Reporting conclusions from a Bayesian analysis

Any Bayesian analysis produces a posterior density, $\pi(\theta|y)$, for a parameter of interest. As well as graphical representations of the $\pi(\theta|y)$ (e.g. a plot the of the posterior density function) we can summarise it by reporting:

- the posterior mean;
- the posterior variance or standard deviation.

In many simple models, the posterior comes from a family whose mean and variance can be written down easily from standard formulae (e.g., given on the AMS formula sheet!).

In addition to the posterior mean and variance, it is often useful to report intervals where $\theta|y$ accumulates most of its posterior mass. More precisely, since $\pi(\theta|y)$ represents our posterior belief about θ having observed y, it is natural to express our posterior belief that θ lies in a particular interval (θ_L, θ_U) as

$$\mathbf{P}\left(\theta \in (\theta_L, \theta_U) \mid y\right) = \int_{\theta_L}^{\theta_U} \pi(\theta|y) d\theta.$$

Now if we select (θ_L, θ_U) so that this probability is $(1-\alpha)$, then (θ_L, θ_U) is what is known as a $100(1-\alpha)\%$ *credible interval* for θ. To construct this interval we simply exclude an area of $\frac{\alpha}{2}$ from each tail of the distribution. The quantities θ_L and θ_U are respectively the $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ quantiles of the posterior distribution.

For instance, in our previous example involving a binomial model, the posterior density for p was Beta(29, 81). We can get an equal tailed 95%-credible interval for p by using R as follows:

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> pL = qbeta(0.025, 29, 81)
> pU = qbeta(0.975, 29, 81)
> pL
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[1] 0.1859729
> pU
[1] 0.3494381
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giving (0.186, 0.349) as the 95% interval.

Moreover, the posterior mean of p is $29/110 = 0.263$ and the posterior standard deviation is 0.042. Appealing to a normal approximation we could get an approximate 95% credible interval by computing the posterior mean $+/-$ 1.96 standard deviations. This results in the interval (0.182, 0.346), which corresponds quite closely with the interval above.

For multivariate models, we typically report the so-called Highest Posterior Density (HPD) $(1-\alpha)$ credible region by identifying a value $\gamma_\alpha > 0$ such that the set $\mathcal{C}_\alpha = \{\theta : \pi(\theta|y) > \gamma_\alpha\}$ accumulates $(1-\alpha)$ of probability mass, i.e.,

$$\int_{\mathcal{C}_\alpha} \pi(\theta|y)d\theta = 1-\alpha.$$

Lastly, notice that credible intervals are not equivalent to confidence intervals. The latter are frequentist objects where the parameter θ is fixed and the *interval* is the random outcome. Therefore, the level of confidence relates to the frequency with which the random interval contains the true parameter over repeated sampling, rather than a probability that conditions on the particular data observed. By contrast the Bayesian approach lends itself very nicely to reporting and interpreting intervals since we are treating θ as a random variable from the outset. We can therefore use $\pi(\theta|y)$ to make probability statements about θ that are expressed in terms of *conditional probabilities* given y.

Selecting priors - Informative priors

Informative priors are prior distributions that are specified in order to model or reflect prior information or beliefs about the parameter of interest.

Example 1: For any given coin, the probability of 'H' denoted by p is close to 0.5, but not exactly equal to 0.5. Suppose that we expect fluctuations around 0.5 of the order of $\pm 10\%$. Before carrying out a coin-tossing experiment we might assign a prior to p, $p \sim \text{Beta}(100, 100)$ to reflect this belief. In this case, the prior mean is 0.5 and the prior standard deviation is 3.5%, so three standard deviations is approximately 10%. This would be example where we use an *informative prior*.

After an experiment in which we observe r 'H' in n tosses, say, we have

$$p|y \sim \text{Beta}(100 + r, 100 + n - r).$$

Clearly r and $n - r$ must be sufficiently large if the mean and variance of the posterior are to look substantially different from those of the prior. (i.e. we need to collect a lot of data to change our strong prior belief).

Example 2: When using a Gamma(α, β) prior for λ in Poi(λ) or Exp(λ), we often set α and β in order to reflect prior beliefs about the prior mean α/β and the prior variance α/β^2.

In fact, it can be shown that the coefficient of variation (= standard deviation/mean) of the gamma prior equals $1/\sqrt{\alpha}$. If we had chosen a small value of α, our prior distribution would have a large variability about its mean.

Selecting priors - Non-informative priors

Vague priors

What do we do if we don't have a lot of prior information? One option is to use a *vague prior* that 'supports' a wide range of values of the parameter and doesn't strongly influence our inferences. Some examples:

- For p in Bin(n, p) we obtain a vague prior by setting $p \sim$ Beta(α, β) where α and β are small.
- For λ in Poi(λ) or Exp(λ) a vague prior for λ is given by $\lambda \sim$ Gamma(α, β) where α and β are small.
- For μ in N(μ, σ^2) where σ^2 is known, a vague prior for μ could be e.g. $\mu \sim \mathrm{N}(0, \gamma^2)$ where γ^2 is large.

Issues with changes of parametrisation

Many statisticians have tried to identify prior distributions that can be used to represent total ignorance about a parameter and that are robust to different ways of parametrising statistical models. The issue with the choice of the parametrisation is illustrated below.

Example. For p in Bin(n, p) it is tempting to think a 'flat' uniform prior represents ignorance, as this is a vague prior that assigns all values of p the same prior density. However, this property is not preserved under 1-1 parameter transformation. For instance, if we're 'totally ignorant' about p then we are 'totally ignorant' about p^2, which also takes values in $[0, 1]$. However, if $p \sim \mathrm{U}(0, 1)$ then

$$p^2 \sim \text{Beta}\left(\frac{1}{2}, 1\right),$$

which is highly asymmetric and concentrates mass on lower values of p^2, and hence very different to a uniform prior $p^2 \sim \text{Beta}(1, 1)$.

This discrepancy is undesirable, as two statisticians analysing the same data and following the same philosophy to specify a non-informative prior would arrive at different conclusions because they use different parametrisations of the likelihood function.

Jeffreys' prior

The Jeffreys' prior is a general strategy for deriving priors that are invariant to changes of parametrisation of the likelihood function. The Jeffreys' prior for an unknown parameter θ is given by

$$\pi_J(\theta) \propto I(\theta)^{\frac{1}{2}},$$

where we recall that $I(\theta)$ denotes the Fisher information. This prior is often improper (cannot be normalised to define a proper density with unit probability mass), and is hence only specified up to a proportionality constant. The construction of the prior stems from:

(i) considering a parameterisation $\phi = h(\theta)$ such that $I(\phi)$ is constant for all values of ϕ.

(ii) assigning a flat prior $\pi(\phi) \propto 1$ to ϕ;

(iii) transforming back to the original parameterisation to obtain the prior on θ.

It is easy to show that the resulting prior is invariant to changes of parametrisation.

Problem with Jeffreys priors: Your choice of prior depends on the particular experiment that you do! To see this, suppose you are totally ignorant about $p = \mathbf{P}(\mathrm{H})$ for a given coin. Consider the following two experiments:

1. Toss the coin n times and count the number of heads.

2. Count the number of tosses required to achieve the first 'H'.

We find different Jeffrey's priors depending on which experiment we choose!

Also, as mentioned previously, Jeffreys' priors are often improper. In some cases, this can lead to invalid posterior distributions and lead to nonsensical inferences.

Location and scale parameters: In the case of *location* parameters on $(-\infty, \infty)$, i.e., where the likelihood $L(\theta; y) = L(0; y - \theta)$ for all $\theta \in (-\infty, \infty)$, then

$$\pi_J(\theta) \propto 1.$$

For example, in $N(\mu, \sigma^2)$ where σ^2 is known, μ is a location parameter.

Remark. The prior $\pi_J(\mu) = 1$ over $(-\infty, \infty)$ is improper, as a result Bayes' theorem does not hold. However, we may still define (not derive) the posterior distribution by mimicking Bayes' theorem, i.e.,

$$\pi(\theta|y) = \frac{\pi(\theta)L(\theta; y)}{\int \pi(\theta')L(\theta'; y)d\theta'}, \qquad (**)$$

which is valid so long as the integral on the denominator is finite (this needs to be checked!).

Similarly, for a *scale* parameter whose range is $(0, \infty)$, i.e., a parameter for which the likelihood function verifies $L(\theta; y) = L(1; \theta^{-1}y)/\theta$ for all $\theta \in (0, \infty)$, then

$$\pi_J(\theta) \propto \frac{1}{\theta}.$$

For instance, the parameters λ in $\text{Exp}(\lambda)$ and σ in $\text{N}(0, \sigma^2)$ are scale parameters. Hence,

$$\pi_J(\lambda) \propto \frac{1}{\lambda},\ \pi_J(\sigma) \propto \frac{1}{\sigma}.$$

Exercises: Derive the Jeffreys' priors for $\text{Bin}(n, p)$, $\text{N}(\mu, \sigma^2)$ with known σ^2, $\text{Exp}(\lambda)$, and $\text{N}(0, \sigma^2)$, respectively.

Predictive distributions

In statistical inference, once we have fitted a model to some observed data, we often want to use the model to make predictions about future outcomes. Bayesian inference allows you to naturally incorporate in your predictions the uncertainty about the unknown parameters of the model. We illustrate this with a simple example.

Example: You are in charge of recruitment in a company. At the start, you have no knowledge of the quality of job applicants from university X. In the first year, you receive 10 applications of which 3 turn out to be suitable. During the second year you hear that you will shortly receive 3 further applications from university X. Let Z denote the (as yet unknown) number of these applicants who will turn out to be suitable. How can you identify a probability distribution to describe your beliefs regarding Z?

Solution: In Bayesian statistics, the solution to this problem is to determine the predictive distribution for Z given the observation. We can proceed as follows:

Let p denote the unknown probability that an applicant from university X is suitable. At the beginning you were ignorant about p. For simplicity, let's suppose you represented that using a uniform prior for p, that is

$$p \sim \text{U}(0, 1) \equiv \text{Beta}(1, 1)\,.$$

During the first year you obtain data y (3 suitable out of 10). Assuming a binomial model and using Bayesian reasoning, you update your prior to obtain a posterior distribution for p:

$$(p|y) \sim \text{Beta}(4, 8).$$

Consider now the distribution of Z. Conditional on p, you know $Z \sim \text{Bin}(3, p)$, but you don't know p. However, you have a distribution for p, provided by the posterior above.

Therefore, by using basic laws of probability, we identify a joint probability function for p and z:

$$\pi(p, z|y) = \pi(p|y)\mathbb{P}(Z = z|p),$$

where $\pi(p|y)$ is the Beta(4, 8) density and $\mathbb{P}(Z = z|p)$ is the binomial probability mass function.

Again, following basic probability laws, we now obtain the marginal distribution for Z by integrating with respect to p. This gives what is known as the predictive probability mass function for Z given observation y, which is essentially the conditional distribution

$$f_Z(z|y) = \pi(z|y) = \int \pi(p, z|y)dp = \int \pi(p|y)\mathbb{P}(Z = z|p)dp.$$

Notice that $f_Z(z|y)$ corresponds to the posterior expectation of the function $g(p) = \mathbb{P}(Z = z|p)$. Therefore we are simply averaging the value of $\mathbb{P}(Z = z|p)$ over the posterior of $p|y$.

Let us calculate this for our example:

$$\begin{aligned} f_Z(z|y) &= \int_{p=0}^{1} \pi(p|y)\mathbb{P}(Z = z|p)dp \\ &= \int_{p=0}^{1} \frac{\Gamma(12)}{\Gamma(4)\Gamma(8)} p^3(1-p)^7 \binom{3}{z} p^z(1-p)^{3-z} dp \\ &= \frac{\Gamma(12)}{\Gamma(4)\Gamma(8)} \binom{3}{z} \int_{p=0}^{1} p^{z+3}(1-p)^{10-z} dp \\ &= \frac{\Gamma(12)}{\Gamma(4)\Gamma(8)} \binom{3}{z} \frac{\Gamma(4+z)\Gamma(11-z)}{\Gamma(15)}. \end{aligned}$$

Let us use this to calculate the predicative probability that none of the candidates will be suitable:

$$\mathbb{P}(Z = 0|X = 3) = \frac{\Gamma(12)}{\Gamma(4)\Gamma(8)} \binom{3}{0} \frac{\Gamma(4)\Gamma(11)}{\Gamma(15)} = 0.3297.$$

The resulting distribution $f_Z(z|y)$ is an example of a beta-binomial distribution. You can think of it as a particular kind of mixture distribution, where a random draw is obtained by first drawing a probability p from a beta distribution, and then drawing a sample from a binomial distribution with that particular value of p. To generate a random sample of size n from it, first draw a random sample $p_1, p_2, \ldots, p_n$ from Beta(4, 8), and then draw $z_i \sim \text{Bin}(3, p_i), i = 1, 2, 3, \ldots n$. Using R you could easily investigate its properties using simulation. However we have an analytic form for the probability mass function which we can compute using R.

Now consider the values of $f_Z(z|y)$ for $z = 0, 1, 2, 3$.

```
> z = c(0:3)
> fz = 6*beta(4+z, 11-z)/(beta(4, 8)*gamma(z+1)*gamma(3-z+1))
> fz
[1] 0.32967033  0.39560440  0.21978022  0.05494505
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Again we see that the predictive probability that we get no suitable candidates is 0.3297.

Now compare what would we would have predicted had we assumed that p was equal to its posterior mean 1/3. (This is not fully Bayesian!)

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> fz2 = dbinom(z, 3, 1/3)
> fz2
[1] 0.29629630 0.44444444 0.22222222 0.03703704
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We find that conditioning on the posterior mean leads to smaller estimates for the probability of extreme values ($z = 0$ and $z = 3$). This illustrates the fact that by taking account of the posterior uncertainty in p, the fully Bayesian approach predicts that extreme events are more likely in comparison to approaches that use a point estimate for p. This tends to hold in general. That is, if you take account of parameter uncertainty when making predictions, you will recognise a greater possibility that extreme events may occur. This is particularly important in applications that require accurately predicting quantiles.

Exercises: Derive the predictive distributions for the gamma-Poisson and gamma-exponential models, respectively.